

Assignment - 1

Q1. $\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{120}{2} = 60 \text{ km/h}$

- The car's position as a function of time is, $s(t)$
It is cont. on the interval $[0, 2]$

- The function $s(t)$ is diff. on the interval $(0, 2)$.

Using mean value theorem:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Here, $a = 0$; $b = 2$

$$s(0) = 0 \text{ km (At point A)}$$

$$s(2) = 120 \text{ km (At point B)}$$

$$\frac{s(2) - s(0)}{2 - 0} = \frac{120 - 0}{2} = 60 \text{ km/h}$$

$\therefore$ There exists some $c \in (0, 2)$, such that

$$s'(c) = 60 \text{ km/h}$$

Q2. Let u and v be functions of x such that n^{th} derivative exists, then the n^{th} derivative of

$$(u \cdot v)^{(n)} = {}^nC_0 u^{(n)} v + {}^nC_1 u^{(n-1)} v' + {}^nC_2 u^{(n-2)} v'' + \dots + {}^nC_{n-1} u' v^{(n-1)} + {}^nC_n u v^{(n)}$$

$$\Rightarrow y^{1/m} + y^{-1/m} = 2x$$

$$\text{Let, } u(x) = y(x)^{1/m}$$

$$\text{Then, } u + \frac{1}{u} = 2x$$

$$u^2 - 2xy + y = 0$$

After differentiation

Differentiating w.r. to x :

$$\frac{1}{m} y^{\frac{1}{m}-1} y_1 - \frac{1}{m} y^{-\frac{1}{m}-1} y_1 = 2$$

$$y^{\frac{1}{m}} y_1 - y^{-\frac{1}{m}} y_1 = 2my$$

[x by my]

$$y_1 (y^{\frac{1}{m}} - y^{-\frac{1}{m}}) = 2my$$

$$\Rightarrow y_1^2 (y^{\frac{1}{m}} - y^{-\frac{1}{m}})^2 = (2my)^2$$

$$y_1^2 (y^{\frac{2}{m}} - 2 + y^{-\frac{2}{m}}) = 4m^2 y^2$$

$$[(y^{\frac{1}{m}} + y^{-\frac{1}{m}})^2 = (2x)^2]$$

$$\therefore y^{\frac{2}{m}} + 2 + y^{-\frac{2}{m}} = 4x^2$$

$$y^{\frac{2}{m}} + y^{-\frac{2}{m}} = 4x^2 - 2$$

put this in y_1^2 : $y_1^2 (4x^2 - 2 - 2) = 4m^2 y^2$

$$y_1^2 (4x^2 - 4) = 4m^2 y^2$$

$$y_1^2 (x^2 - 1) = m^2 y^2$$

Now, Diff. w.r. to x

$$2y_1 y_2 (x^2 - 1) + y_1^2 (2x) = 2m^2 y y_1$$

$$\Rightarrow (x^2 - 1) y_2 + x y_1 = m^2 y$$

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Use Leibniz's to find the n -th derivative

$$(x^2 - 1)y_2 + xy_1 - m^2 y = 0$$

For 1st term, $(x^2 - 1)y_z$

1. $u = y_2$; $v = x^2 - 1 \cdot 4^{(n-k)}$

$$u = y_{n+2} - k v$$

$$= x^{k-1} v' = 2x^k v = 0 \quad ; k \geq 3$$

Using Leibnitz thm :

$$D^n((x^2-1)y_2) = \binom{n}{0} y_{n+2}(x^2-1) + \binom{n}{1} y_{n+1}(2x) + \binom{n}{2} y_n(2)$$

$$D^n((x^2-1)y_2) = (x^2-1)y_{n+2} + 2nxy_{n+1} + n(n-1)y_n$$

- For 2nd term, xy

$$u, \quad u = y, \quad v = x \cdot u^{(n-k)}$$

Using Him:

$$D^n(x y_i) = \binom{n}{0} y_{n+1}(x) + \binom{n}{1} y_n(x)$$

$$D^n(x y_1) = x y_{n+1} + n y_n x'$$

- For IIIrd Term, $-m^2 y$

$$1) \quad (-m^2 y) = -m^2 y$$

Now,

$$(x^2-1)y_{n+2} + (2nx+x)y_{n+1} + (n(n-1)+n-m^2)y_n = 0$$

$$(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-n+m^2)y_n = 0$$

$$(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0$$

Q3. $P(x, y) = 40x + 50y - (2x^2 + 2y^2 + xy)$

$$P_x = 40 - 4x - y = 0 \Rightarrow 4x + y = 40$$

$$P_y = 50 - x - 4y = 0 \Rightarrow x + 4y = 50$$

From, $4x + y = 40$

$$y = 40 - 4x$$

put y in $x + 4y = 50$,

$$x + 4(40 - 4x) = 50$$

$$x + 160 - 16x = 50$$

$$-15x = -110 \Rightarrow x = \frac{110}{15}$$

$$x = \frac{110}{15} = \frac{22}{3}$$

Note,

$$y = 40 - 4 \times \frac{22}{3}$$

$$= 40 - \frac{88}{3}$$

$$= \frac{120 - 88}{3}$$

$$= \frac{32}{3}$$

$$\text{So } (x, y) = \left(\frac{22}{3}, \frac{32}{3} \right)$$

$$H = \begin{pmatrix} P_{xx} & P_{xy} \\ P_{yx} & P_{yy} \end{pmatrix} = \begin{pmatrix} -4 & -1 \\ -1 & -4 \end{pmatrix}$$

$$P_{xx} = -4 < 0; \quad P_{yy} = -4 < 0; \quad (P_{xy} = P_{yx}) = -1$$

$$(-4)(-4) - (-1)^2 = 16 - 1 = 15 > 0$$

$$P_{xx} = -4 < 0$$

Maximum Profit $40x + 50y = 2480$

$$x = \frac{22}{3}, \quad y = \frac{32}{3}$$

$$40x + 50y = \frac{2480}{3}$$

$$2x^2 + 2y^2 + xy = \frac{1240}{3}$$

$$P_{\max} = \frac{2480}{3} = \frac{1240}{3}$$

Now,

$$P_{\max} = \frac{1240}{3} \approx 413.333 \text{ thousand}$$

$$= ₹ 413,333.33$$

34. A function $u(x, y)$ is called a homogeneous function of degree n if, $u(\lambda x, \lambda y) = \lambda^n u(x, y)$
 $\lambda \in \mathbb{R}, \lambda > 0$

If $u(x, y)$ is a homogeneous function of x & y of degree n , then $x \frac{du}{dx} + y \frac{du}{dy} = nu$

Given, $u(x, y) = y^2 e^{\frac{x}{y}} + x^2 \tan^{-1}\left(\frac{x}{y}\right)$

Now, $u(tx, ty) = (ty)^2 e^{\frac{tx}{ty}} + (tx)^2 \tan^{-1}\left(\frac{tx}{ty}\right)$

$$= t^2 y^2 e^{\frac{x}{y}} + t^2 x^2 \tan^{-1}\left(\frac{x}{y}\right)$$

$$= t^2 u(x, y)$$

$\therefore$ It is a homogeneous function of degree $n=2$

Since, $u(x,y)$ is a homo. function of degree 2.
Then, $x \frac{du}{dx} + y \frac{du}{dy} = 2u$

Using Euler's Thm,

$$x^2 \frac{d^2 u}{dx^2} + 2xy \frac{d^2 u}{dx dy} + y^2 \frac{d^2 u}{dy^2} = n(n-1)u$$

Now, $n=2$

$$x^2 \frac{d^2 u}{dx^2} + 2xy \frac{d^2 u}{dx dy} + y^2 \frac{d^2 u}{dy^2} = 2u$$

35. Let $u(x,y)$, $v(x,y)$ be function of x & y
The Jacobian of u, v w.r.t. x, y is

$$J(u,v) = \begin{vmatrix} \frac{du}{dx} & \frac{du}{dy} \\ \frac{dv}{dx} & \frac{dv}{dy} \end{vmatrix}$$

When $m=n$ (square Jacobian), the Jacobian determinant $\det J$ or $\det \left(\frac{d(F_1, \dots, F_n)}{d(x_1, \dots, x_n)} \right)$ is the determinant of that matrix.

Four common applications:

- 1) Change of variables in multiple integrals
- 2) Inverse function theorem
- 3) Newton's method for systems
- 4) Stability / sensitivity analysis

Given, $u = x^2 - y^2$; $x = r \cos \theta$
 $v = 2xy$; $y = r \sin \theta$

$$\begin{aligned}u &= (r \cos \theta)^2 - (r \sin \theta)^2 \\&= r^2 \cos^2 \theta - r^2 \sin^2 \theta \\&= r^2 (\cos^2 \theta - \sin^2 \theta) = r^2 \cos(2\theta)\end{aligned}$$

$$\begin{aligned}v &= 2(r \cos \theta)(r \sin \theta) \\&= 2r^2 \cos \theta \sin \theta \\&= r^2 (2 \cos \theta \sin \theta) = r^2 \sin(2\theta)\end{aligned}$$

$$\frac{du}{dr} = \frac{d}{dr} (r^2 \cos(2\theta)) = 2r \cos(2\theta)$$

$$\frac{du}{d\theta} = \frac{d}{d\theta} (r^2 \cos(2\theta)) = -2r^2 \sin(2\theta)$$

$$\frac{dv}{dr} = \frac{d}{dr} (r^2 \sin(2\theta)) = 2r \sin(2\theta)$$

$$\frac{dv}{d\theta} = \frac{d}{d\theta} (r^2 \sin(2\theta)) = 2r^2 \cos(2\theta)$$

$$J = \begin{pmatrix} \frac{du}{dr} & \frac{du}{d\theta} \\ \frac{dv}{dr} & \frac{dv}{d\theta} \end{pmatrix} = \begin{pmatrix} 2r \cos(2\theta) & -2r^2 \sin(2\theta) \\ 2r \sin(2\theta) & 2r^2 \cos(2\theta) \end{pmatrix}$$

$$\begin{aligned}\left| \frac{d(u,v)}{d(r,\theta)} \right| &= (2r \cos(2\theta))(2r^2 \cos(2\theta)) - (-2r^2 \sin(2\theta))(2r \sin(2\theta)) \\&= 4r^3 \cos^2(2\theta) + 4r^3 \sin^2(2\theta) \\&= 4r^3 (\cos^2(2\theta) + \sin^2(2\theta)) \\&= 4r^3\end{aligned}$$

Q6. $T(x, y) = x^2 + y^2$
 $x = r \cos \theta$
 $y = r \sin \theta$

$$\begin{aligned} T(r, \theta) &= (r \cos \theta)^2 + (r \sin \theta)^2 \\ &= r^2 \cos^2 \theta + r^2 \sin^2 \theta \\ &= r^2 (\cos^2 \theta + \sin^2 \theta) = r^2 \end{aligned}$$

The circular plate has a radius of 2.
 i.e., $0 \leq r \leq 2$

For a full circle, the angle θ ranges from 0 to 2π
 i.e., $0 \leq \theta \leq 2\pi$

$$\iint_R T(x, y) dA, \quad dA = r dr d\theta$$

$$\int_0^{2\pi} \int_0^2 r^2 \cdot r dr d\theta = \int_0^{2\pi} \int_0^2 r^3 dr d\theta$$

$$\Rightarrow \int_0^2 r^3 dr = \left[\frac{r^4}{4} \right]_0^2 = \frac{2^4}{4} - \frac{0^4}{4} = 4$$

$$\Rightarrow \int_0^{2\pi} 4 d\theta = [4\theta]_0^{2\pi} = 4(2\pi) = 8\pi$$

The area of the plate,

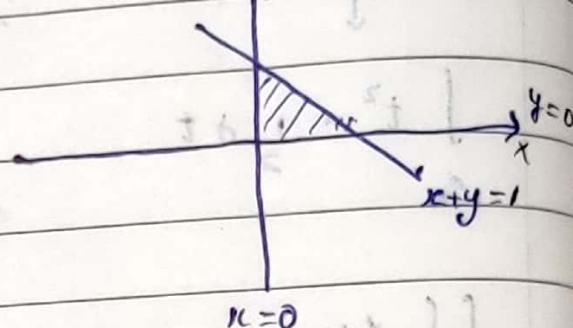
$$A = \pi R^2$$

$$= \pi (2)^2 = 4\pi$$

$$T_{\text{avg}} = \frac{\text{Integral of } T(x, y) \text{ over the plate}}{\text{Area of the plate}} = \frac{8\pi}{4\pi} = 2$$

$\therefore$ The avg. temp. of the plate is 2°C .

$$\begin{cases} x-y=4 \\ x+y=v \end{cases} \Rightarrow x = \frac{u+v}{2} ; y = \frac{v-u}{2}$$



$$\iint_R \cos\left(\frac{x-y}{x+y}\right) dx dy$$

$$= \iint_{R'} \cos\left(\frac{u}{v}\right) |J| du dv$$

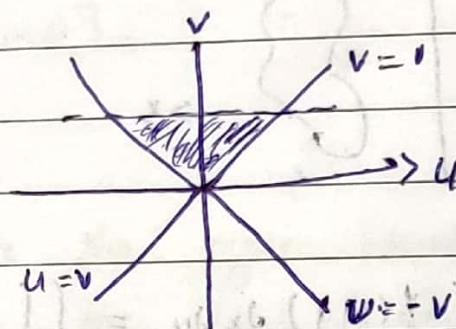
Now,

$$|J| = \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{2}$$

$$J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{\frac{\partial(u,v)}{\partial(x,y)}}$$

$$= \iint_{R'} \cos\left(\frac{u}{v}\right) \times \frac{1}{2} du dv$$



R	R'
$x=0$	$u=-v$
$y=0$	$v=u$
$x+y=1$	$v=1$

$$x-y=4 \Leftrightarrow x = \frac{u+v}{2} \Rightarrow x=0 \Rightarrow u+v=0 \Rightarrow u=-v$$

$$x+y=v \Leftrightarrow y = \frac{v-u}{2} \Rightarrow y=0 \Rightarrow v-u=0 \Rightarrow v=u$$

$$x \geq 0, \quad \frac{u+v}{2} \geq 0 \Rightarrow u \geq -v$$

$$y \geq 0, \quad \frac{v-u}{2} \geq 0 \Rightarrow v \geq u$$

$$x+y \leq 1, \quad v \leq 1$$

$$\Rightarrow \iint_R \cos\left(\frac{y}{v}\right) \frac{1}{2} du dv$$

$$= \frac{1}{2} \int_{v=0}^1 \int_{u=-v}^v \cos\left(\frac{y}{v}\right) du dv$$

$$= \frac{1}{2} \int_{v=0}^1 \left[\frac{\sin\left(\frac{y}{v}\right)}{1/v} \right]_{u=-v}^v dv$$

$$= \frac{1}{2} \int_{v=0}^1 v [\sin(1) - \sin(-1)] dv$$

$$= \frac{1}{2} \int_0^1 2v \sin 1 dv = \sin 1 \int_0^1 v dv = \sin 1 \cdot \frac{1}{2} = \frac{\sin 1}{2}$$

28.

$$xy = 4 ; xy = 16$$

$$yz = 1 ; yz = 4$$

$$zx = 25 ; zx = 49$$

$$\text{Let, } u = xy$$

$$v = yz$$

$$w = zx$$

$$\Rightarrow 4 \leq u \leq 16$$

$$\Rightarrow 1 \leq v \leq 4$$

$$\Rightarrow 25 \leq w \leq 49$$

$$\text{Now, } V = \iiint_R dx dy dz$$

$$V = \int_{25}^{49} \int_1^4 \int_4^{16} \left| \frac{d(x,y,z)}{d(u,v,w)} \right| du dv dw$$

$$\left| \frac{d(x,y,z)}{d(u,v,w)} \right| = \frac{1}{2\sqrt{uvw}}$$

$$x = \sqrt{\frac{uw}{v}} ; y = \sqrt{\frac{uv}{w}}$$

$$z = \sqrt{\frac{vw}{u}}$$

$$\Rightarrow V = \int_{25}^{49} \int_1^4 \int_4^{16} \frac{1}{2\sqrt{uvw}} du dv dw$$

$$V = \frac{1}{2} \left(\int_4^{16} \frac{1}{\sqrt{u}} du \right) \left(\int_1^4 \frac{1}{\sqrt{v}} dv \right) \left(\int_{25}^{49} \frac{1}{\sqrt{w}} dw \right)$$

$$\Rightarrow \int_4^{16} u^{-1/2} du = [2\sqrt{u}]_4^{16}$$

$$= (2\sqrt{16}) - (2\sqrt{4})$$

$$= 8 - 4 = 4$$

$$\Rightarrow \int_1^4 v^{-1/2} dv = \left[2\sqrt{v} \right]_1^4$$

$$= (2\sqrt{4}) - (2\sqrt{1})$$

$$= 4 - 2 = 2$$

$$\Rightarrow \int_{25}^{49} w^{-1/2} dw = \left[2\sqrt{w} \right]_{25}^{49}$$

$$= (2\sqrt{49}) - (2\sqrt{25})$$

$$= 14 - 10 = 4$$

Now, $V = \frac{1}{2} \times 4 \times 2 \times 4 = 16$ (cubic units)

Q9. $z = \sqrt{2(x^2 + y^2)}$
 $x^2 + y^2 + z^2 = 9$

Let, $x = \rho \sin \phi \cos \theta$
 $y = \rho \sin \phi \sin \theta$
 $z = \rho \cos \phi$

Let Sphere: $\rho = 3$ [$\because x^2 + y^2 + z^2 = 9$]
 Cone: $z = \sqrt{2} \sqrt{x^2 + y^2}$

So,

$$\rho \cos \phi = \sqrt{2} \rho \sin \phi \Rightarrow \cot \phi = \sqrt{2}$$

The cone corresponds to the polar angle $\phi = \phi_0$
 so satisfying $\tan \phi_0 = \frac{1}{\sqrt{2}}$

$$\cos \phi_0 = \frac{1}{\sqrt{1+\tan^2 \phi_0}} = \frac{1}{\sqrt{1+1/2}} = \frac{1}{\sqrt{3/2}} = \sqrt{\frac{2}{3}}$$

Vol. of the ice-cream (inside the sphere $\rho \leq 3$ and $0 \leq \phi \leq \phi_0$, $0 \leq \theta \leq 2\pi$)

$$V = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\phi_0} \int_{\rho=0}^3 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$
$$= 2\pi \left(\int_0^{\phi_0} \sin \phi \, d\phi \right) \left(\int_0^3 \rho^2 \, d\rho \right)$$

$$\Rightarrow \int_0^3 \rho^2 \, d\rho = \frac{3^3}{3} = 9$$

$$\Rightarrow \int_0^{\phi_0} \sin \phi \, d\phi = 1 - \cos \phi_0$$
$$= 1 - \sqrt{\frac{2}{3}}$$

$$\text{Now, } V = 2\pi \cdot 9 \cdot (1 - \sqrt{2/3})$$

$$= 18\pi (1 - \sqrt{2/3})$$

$$V = 6\pi (3 - \sqrt{6})$$

$$\Rightarrow V \approx 18\pi (1 - \sqrt{2/3})$$

$$Q10. CP) \int_0^1 x^3 (1-\sqrt{x})^5 dx$$

$$du, t = \sqrt{x}$$

$$x = t^2$$

$$dx = 2t dt$$

$$\text{When } x=0, t=0$$

$$\text{When } x=1, t=1$$

$$= \int_0^1 (t^2)^3 (1-t)^5 (2t dt)$$

$$= \int_0^1 t^6 (1-t)^5 (2t) dt$$

$$= 2 \int_0^1 t^7 (1-t)^5 dt$$

$$\text{Compare with } 2 \int_0^1 t^{m-1} (1-t)^{n-1} dt$$

$$= 2 B(m, n)$$

$$\rightarrow m-1 = 7$$

$$\Rightarrow m = 8$$

$$\rightarrow n-1 = 5$$

$$\Rightarrow n = 6$$

$$\therefore 2 B(8, 6)$$

$$I = 2 \frac{\sqrt{8} \sqrt{6}}{\sqrt{8+6}} = 2 \frac{\sqrt{8} \sqrt{6}}{\sqrt{14}}$$

$$= 2 \times \frac{7! \cdot 5!}{13!}$$

$$I = \frac{2 \times 5!}{13 \times 12 \times 11 \times 10 \times 9 \times 8} = \frac{2 \times 120}{(12 \times 10) \times 13 \times 11 \times 9 \times 8}$$

$$= \frac{1}{13 \times 11 \times 9 \times 4}$$

$$= \frac{1}{5148}$$

(99) $\int_0^{\pi} x \sin^7 x \cos^4 x \, dx$

Using, $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$

Let, $a = \pi$

$$I = \int_0^{\pi} (\pi - x) \sin^7(\pi - x) \cos^4(\pi - x) \, dx$$

$$= \int_0^{\pi} (\pi - x) (\sin x)^7 (-\cos x)^4 \, dx$$

$$= \int_0^{\pi} (\pi - x) \sin^7 x \cos^4 x \, dx$$

$$= \pi \int_0^{\pi} \sin^7 x \cos^4 x \, dx - \int_0^{\pi} x \sin^7 x \cos^4 x \, dx$$

$$I = \pi \int_0^{\pi} \sin^7 x \cos^4 x \, dx - I$$

$$2I = \pi \int_0^{\pi} \sin^7 x \cos^4 x \, dx$$

$$\text{Using } \int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$$

$$\text{Let } 2a = \pi$$

$$= 2 \int_0^{\pi/2} \sin^7 x \cos^4 x dx$$

$$2I = \pi \left(2 \int_0^{\pi/2} \sin^7 x \cos^4 x dx \right)$$

$$I = \pi \int_0^{\pi/2} \sin^7 x \cos^4 x dx$$

$$\text{Now, } I = \pi \left[\frac{\left(\frac{7+1}{2}\right) \left(\frac{4+1}{2}\right)}{2 \left(\frac{7+4+2}{2}\right)} \right] = \pi \left[\frac{4 \cdot 5}{2 \sqrt{13/2}} \right]$$

$$= \pi \left[\frac{6 \sqrt{5/2}}{2 \cdot \left(\frac{1}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{5}{2} \sqrt{5/2}\right)} \right]$$

$$= \pi \left[\frac{6}{3405/8} \right] = \frac{48\pi}{3405}$$

$$1155$$

$$(iii) \int_0^{\pi/4} \cos^3(2\theta) \sin^4(4\theta) d\theta$$

$$\sin^4(4\theta) = 16 \sin^4(2\theta) \cos^4(2\theta)$$

$$I = \int_0^{\pi/4} \cos^3(2\theta) \cdot 16 \sin^4(2\theta) \cos^4(2\theta) d\theta$$

$$= 16 \int_0^{\pi/4} \sin^4(2\theta) \cos^7(2\theta) d\theta$$

$$\text{Let, } x = 2\theta$$

$$dx = 2 d\theta \quad ; \quad d\theta = dx/2$$

$$\text{When } \theta = 0, \quad x = 0$$

$$\text{When } \theta = \pi/4, \quad x = \pi/2$$

$$I = 8 \int_0^{\pi/2} \sin^4(x) \cos^7(x) dx$$

$$I = 8 \left[\frac{\sqrt{\frac{4+1}{2}} \sqrt{\frac{7+1}{2}}}{2 \sqrt{\frac{4+7+2}{2}}} \right] = 4 \frac{\sqrt{5/2} \sqrt{4}}{\sqrt{13/2}}$$

$$I = 4 \times \frac{\sqrt{5/2} \cdot 6}{\frac{11 \times 9 \times 7 \times 5}{16} \sqrt{5/2}} = \frac{24 \times 16}{3465} = \frac{384}{3465}$$

$$I = \frac{128}{1155}$$